

Norway

SLEIPNER

Volve F

F-9

F-9 A

Design: F-9 A

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-9	North Reference:	Grid
Wellbore:	F-9 A	Survey Calculation Method:	Minimum Curvature
Design:	F-9 A	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9				
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N	
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E	
Position Uncertainty:		0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-9					
Well Position	+N/-S	-1.96 m	Northing:	6,478,561.57 m	Latitude:	58° 26' 29.7442 N
	+E/-W	1.11 m	Easting:	435,051.13 m	Longitude:	1° 53' 14.9992 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	F-9 A				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2009	27.06.2009	-1.99	71.60	50,401

Design	F-9 A				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	420.40
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	-1.96	1.11	190.04	

Survey Program	Date	27.09.2010			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
218.50	420.40	F9 17 1/2in Surveys (F-9)	Keeper, stat	Gyro Tool from SDC	
454.90	537.60	F-9 A 12 1/4" GWD (F-9 A)	Keeper, stat	Gyro Tool from SDC	
565.68	968.13	F-9 A 12 14" MWD (F-9 A)	Magnetic, std, mag-corr	Magnetic Tools (MWD, EMS)	
968.13	1,181.03	F-9 A 8 1/2" MWD (F-9 A)	Magnetic, std, mag-corr	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-1.96	1.11	0.00	0.000	0.000	0.000	
218.50	0.26	197.21	218.50	-2.11	1.07	0.16	0.107	0.107	0.000	
259.00	0.18	255.31	259.00	-2.22	0.98	0.28	0.167	-0.059	43.037	
299.50	0.35	240.89	299.50	-2.29	0.81	0.39	0.134	0.126	-10.681	
340.00	0.44	34.96	340.00	-2.23	0.79	0.32	0.570	0.067	114.126	
394.30	0.35	344.16	394.30	-1.90	0.86	-0.02	0.193	-0.050	-28.066	
420.40	0.35	352.69	420.40	-1.74	0.83	-0.16	0.060	0.000	9.805	
454.90	4.66	197.74	454.86	-2.97	0.39	1.13	4.330	3.748	-134.739	

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Wellbore:	F-9 A	Survey Calculation Method:	Minimum Curvature
Design:	F-9 A	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
497.70	8.93	199.58	497.35	-7.76	-1.25	6.13	2.996	2.993	1.290
537.60	14.03	193.34	536.44	-15.39	-3.41	14.02	3.941	3.835	-4.692
565.68	17.50	186.45	563.46	-22.90	-4.67	21.63	4.206	3.707	-7.361
604.93	21.91	183.39	600.41	-36.08	-5.77	34.80	3.461	3.371	-2.339
686.26	28.66	182.17	673.90	-70.76	-7.40	69.23	2.497	2.490	-0.450
727.00	30.48	183.06	709.34	-90.84	-8.33	89.17	1.379	1.340	0.655
767.66	33.77	185.48	743.77	-112.39	-9.96	110.67	2.606	2.427	1.786
807.83	37.83	187.45	776.34	-135.73	-12.62	134.12	3.152	3.032	1.471
847.97	41.28	189.89	807.29	-160.99	-16.49	159.67	2.828	2.578	1.824
888.00	46.52	189.17	836.12	-188.35	-21.08	187.41	3.945	3.927	-0.540
928.41	50.33	189.21	862.93	-218.19	-25.90	217.63	2.829	2.829	0.030
968.13	54.27	190.58	887.22	-249.14	-31.31	249.05	3.086	2.976	1.035
1,059.44	59.07	191.08	937.37	-324.05	-45.65	325.32	1.583	1.577	0.164
1,140.67	58.96	190.82	979.19	-392.42	-58.88	394.95	0.092	-0.041	-0.096
1,181.03	58.83	190.55	1,000.04	-426.38	-65.29	429.50	0.197	-0.097	-0.201
1,206.00	58.83	190.55	1,012.97	-447.38	-69.20	450.86	0.000	0.000	0.000

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
- hit/miss target									
- Shape									
F-9A Geological Target 2	1.00	1.00	887.00	-246.18	-42.29	6,478,317.43	435,007.75	58° 26' 21.8291 N	1° 53' 12.5737 E
- actual wellpath misses target center by 11.33m at 967.18m MD (886.66 TVD, -248.39 N, -31.17 E)									
- Polygon									
Point 1			882.80	245.12	7.57	6,478,077.47	434,957.75		
Point 2			887.33	-11.02	52.73	6,478,337.43	434,957.75		
Point 3			887.36	-28.39	-45.78	6,478,337.43	435,057.75		
Point 4			882.83	227.75	-90.95	6,478,077.47	435,057.75		
F-9A Drillers Target 2306	1.00	1.00	887.00	-246.18	-42.29	6,478,317.43	435,007.75	58° 26' 21.8291 N	1° 53' 12.5737 E
- actual wellpath misses target center by 11.33m at 967.18m MD (886.66 TVD, -248.39 N, -31.17 E)									
- Polygon									
Point 1			883.01	230.86	-2.10	6,478,089.83	434,969.74		
Point 2			887.12	-0.92	38.77	6,478,325.06	434,969.74		
Point 3			887.14	-14.13	-36.12	6,478,325.07	435,045.76		
Point 4			883.04	217.65	-76.99	6,478,089.83	435,045.76		

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
220.80	220.80	30"	30.000	36.000	
421.50	421.50	13 3/8" Window	13.375	17.500	
996.00	903.20	9 5/8" liner	4.500	7.000	
1,201.00	1,010.38	6 5/8" liner	4.500	7.000	

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Well:	F-9	North Reference:	Grid
Wellbore:	F-9 A	Survey Calculation Method:	Minimum Curvature
Design:	F-9 A	Database:	Production EDM P246N

Checked By: _____	Approved By: _____	Date: _____
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